

Jesus Rodriguez

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EDUCATION

Brown Univserity

Bachelor of Science in Computer Science

Providence, RI

September 2022 - May 2026

EXPERIENCE

Software Engineer II Intern

Walmart Global Tech

Jun. 2025 - Aug. 2025

Bentonville, AR

- Designed and implemented **FastAPI** endpoints to query and preprocess **CosmosDB SQL** data, aggregate changes, and send decision tree prompts to **GPT-4o**, reducing token usage 1,200x for real-time AI summaries.
- Developed a responsive **React** frontend to visualize AI-generated change summaries in Walmart's Assortment Planning Tool, improving collaboration and decision-making across large-scale plans, **saving 14,800 hours**

Undergraduate Research Assistant

Singh Lab, Brown University

May 2025 – Jan. 2026

Providence, RI

- Adapted and enhanced the **scBasset CNN** architecture using TensorFlow and R to predict cell age from single-cell ATAC-seq embeddings, improving model relevance for genomic age analysis.
- Developed **Python** preprocessing pipelines to clean and transform genomic data, enabling fine-tuning of neural networks for improved prediction accuracy in cell age estimation.

Technical Director

Full Stack at Brown

Feb. 2024 – Present

Providence, RI

- Oversee three full-stack development teams** (14 developers and designers each) and their project managers, guiding technology decisions and ensuring progress through weekly PM meetings.
- Assist developer onboarding**, environment setup, and technical support across teams, resolving issues and maintaining efficient development workflows.

Undergraduate Research Assistant

Anna Aizer, Brown University

Sep. 2025 – Jan. 2026

Providence, RI

- Designed an **AI-assisted OCR pipeline** to digitize historical U.S. Census records, leveraging machine learning and text recognition techniques to automate data extraction and enhance accessibility for socioeconomic research.

Web Designer/Developer and Science Communicator

Serre Lab, Brown University

Mar. 2024 – Sep. 2025

Providence, RI

- Redesigned Serre Lab's website with **Figma** and **React**, enhancing UX and visual appeal for improved engagement.
- Created educational content on neuro-AI for social media, communicating complex concepts to a broad audience.

Data Science Intern, High Performance Computing Team

Center for Computation and Visualization, Brown University

Jun. 2024 – Aug. 2024

Providence, RI

- Deployed a **Docker** container for **NVIDIA's Parabricks** on a **high-performance computing (HPC)** cluster.
- Enhanced performance by leveraging GPU clusters with Parabricks, **reducing runtime from 3 hours to 30 minutes**.

PROJECTS

Deep Learning for Interpretability in Social Science | *TensorFlow, Python*

2024

- Applied deep learning with **TensorFlow** to build control vectors and compared models using causal inference methods like **Double Machine Learning** to evaluate interpretability and performance.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, TypeScript, Java, JavaScript, HTML/CSS, Assembly, Go, R, SQL (Postgres)

Frameworks & Libraries: React, Node.js, Flask, FastAPI, Express PyTorch, TensorFlow, pandas, NumPy, Matplotlib

Developer Tools & Platforms: Git, Docker, TravisCI, AWS, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, CoLab, CosmosDB, Parabricks, Google Cloud Platform

Coursework: Distributed Systems, Cryptography, Deep Learning, Machine Learning, Computational Cognitive Science, Operating Systems, Computer Networks, Data Structures and Algorithms, Programming Languages, Discrete Structures, Computer Systems